MODULE 03

Data Visualization

A concise 10-page starter workbook for Data Science students

Prepared for: V Sai Dhanush

Branch: Data Science | **College:** Mohan Babu University

Learning goal

This workbook introduces the core ideas in data visualization, with a topic-by-topic study path, practical checkpoints, and reflection prompts. Use it as a starting point and supplement it with your class notes and recommended references.

Note: These are sample learning materials created for this demo portal, not official university course notes.

Choosing the Right Chart

Concept overview

In this section, focus on the key principles behind **choosing the right chart**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain choosing the right chart to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Matplotlib Fundamentals

Concept overview

In this section, focus on the key principles behind **matplotlib fundamentals**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain matplotlib fundamentals to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Line Charts and Trend Analysis

Concept overview

In this section, focus on the key principles behind **line charts and trend analysis**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain line charts and trend analysis to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Bar Charts and Comparisons

Concept overview

In this section, focus on the key principles behind **bar charts and comparisons**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain bar charts and comparisons to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Histograms and Distribution Plots

Concept overview

In this section, focus on the key principles behind **histograms and distribution plots**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain histograms and distribution plots to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Scatter Plots and Relationships

Concept overview

In this section, focus on the key principles behind **scatter plots and relationships**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain scatter plots and relationships to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Seaborn for Statistical Graphics

Concept overview

In this section, focus on the key principles behind **seaborn for statistical graphics**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain seaborn for statistical graphics to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Dashboards and Storytelling

Concept overview

In this section, focus on the key principles behind **dashboards and storytelling**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain dashboards and storytelling to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Mini Project: Visual Analytics Report

Concept overview

In this section, focus on the key principles behind **mini project: visual analytics report**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain mini project: visual analytics report to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?